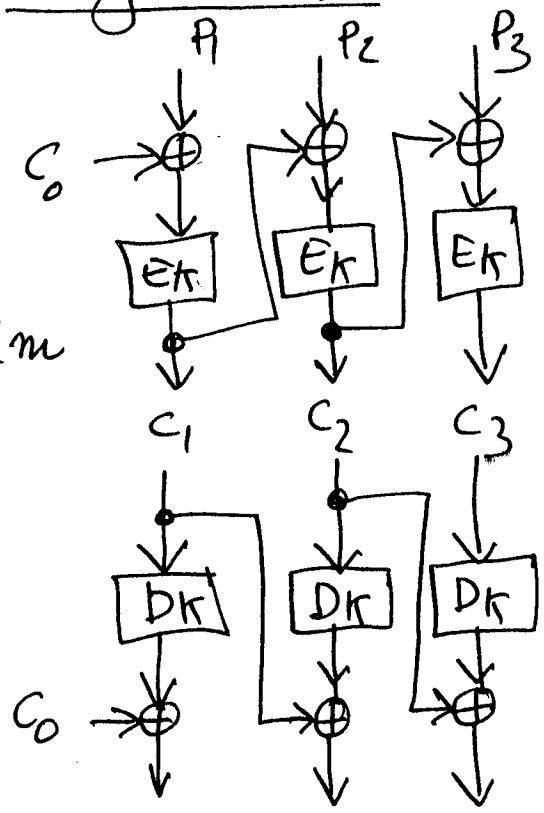
QUESITO 1 — Recupero 'Sicurezza delle Reti' Luglio 2003

(a) encrypt

$c_i = E_k(c_{i-1} \oplus p_i)$

$E_k(x) = (ax+b) \bmod m$



decrypt

$p_i' = c_{i-1} \oplus D_k(c_i)$

$D_k(x) = \bar{a}^1(x-b) \bmod m$

(b) $m = 16$

$C_0 = 1010 =$

$a = 5$

$b = 3$

Affinché $\exists \bar{a}^1$ deve essere $mcd(16,5) = 1$ $a \perp m$ ok

$\bar{a}^{-1} = a^{\varphi(m)-1} \bmod m = 5^7 \bmod 16 = 13$

infatti: $\varphi(m) = 2^4 - 2^3 = 2^3 = 8$

$7 \to 111$

e inoltre:

SQM

1	$1^2 \times 5 = 5$
1	$25 \times 5 = 125 = 13$
1	$13^2 \times 5 = 845 = \underline{13}$

} mod 16

(c) $K \equiv (a, b, IV)$ (2)

$$|K| = N_{chiavi} = m\,\varphi(m)\,2^4 = 16 \times 8 \times 16 = 2048$$

(d) cifratura

$P_1 = 12$; $P_2 = 15$; $P_3 = 0$

$P_1 \equiv 1100 = 'C'$ hex

$P_2 \equiv 1111 = 'F'$ hex

$P_3 \equiv 0000 = '0'$ hex

- $C_1 = E_k(P_1 \oplus IV)$

$$\begin{array}{r} 1100 \\ 1010 \\ \hline 0110 \end{array} \rightarrow 6 \text{ hex}$$

$C_1 \equiv 5 \times 6 + 3 \equiv 33 \bmod 16 = 1_{hex} \equiv 0001$

- $C_2 = E_k(P_2 \oplus C_1)$

$$\begin{array}{r} 1111 \\ 0001 \\ \hline 1110 \end{array} \rightarrow 'D' \equiv 14$$

$C_2 = 5 \times 14 + 3 = 73 \bmod 16 = 9_{hex} \equiv 1001$

- $C_3 = E_k(P_3 \oplus C_2)$

$$\begin{array}{r} 0000 \\ 1001 \\ \hline 1001 \end{array} \rightarrow 9_{hex}$$

$C_3 = 5 \times 9 + 3 \equiv 48 \bmod 16 = 0_{hex}$

$C_1 C_2 C_3 \equiv 190$

(e) decifratura

- $P_1 = D_k(C_1) \oplus IV$

$D_k(1) = 13(1-3) = -26 \bmod 16 = 6 \equiv 0110$

$$\begin{array}{r} 0110 \\ 1010 \\ \hline 1100 \end{array} \rightarrow 12 \hookrightarrow 'C'$$

$P_1 = 1100 = 'C'$ hex

- $P_2 = D_k(C_2) \oplus C_1 = 1111 = 'F'$ hex
- $P_3 = D_k(C_3) \oplus C_2 = 0000 = '0'$ hex

ok

QUESITO 2 Recupero Sicurezza Luglio 2003

(a) p primo ; $0 \le a \le p-2$; $k \in \mathbb{Z}_{p-1}^* \equiv k \perp (p-1)$

(i) i valori di P sono $P \in \mathcal{P} \equiv \mathbb{Z}_p^* \equiv \{1,2,3,4 \dots 52\}$

(ii) $1 \le k \le 51$

essendo $k \in \mathbb{Z}_{p-1}^*$ devo scegliere solo i valori di $k \perp (p-1)$ e cioè $k \perp 52$. Analizzando a uno a uno (sappiamo che $|\mathbb{Z}_{p-1}^*| = \varphi(p-1) = 24$) si ottiene

$k = \{1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 41, 43, 45, 47, 49, 51\}$ 24 termini

(b) Gli elementi primitivi di $\mathbb{Z}_p^*$ sono nel numero di:

$$\#^{\text{elementi primitivi}}_{\mathbb{Z}_p^*} = \varphi(p-1) = \varphi(52) = (13-1)(2^2-2) = 24$$

(c) α è primitivo di $\mathbb{Z}_p^*$ se $\alpha^{\frac{p-1}{q_i}} \bmod p \neq 1 \quad \forall i$

$\alpha = 5$

$p-1 = 52 = 13 \times 2^2$

$5^{26} \bmod 53 = \left(5^{13} \bmod 53\right)^2 \bmod 53 = 52$ OK

$5^4 \bmod 53 = 42$ OK

OK - $\alpha = 5$ è primitivo

(d) $\beta = \alpha^a \bmod p = 5^{19} \bmod 53 = 35$

$19 \to 10011$

1	$12 \times 5 \equiv 5$
0	$25 \equiv 25$
0	$(25^2) \equiv 625 = 42$ mod 53
1	$(42)^2 \times 5 = 8820 = 22$
1	$(22)^2 \times 5 = 2420 = \underline{35}$

(e) $SIG_K(P, k) = (\gamma, \delta)$ $\begin{cases} \gamma = \alpha^k \bmod p \\ \delta = [(P - a\gamma)k^{-1}] \bmod (p-1) \end{cases}$

$k = 1$

$k^{-1} \bmod (p-1) = 1$

$\begin{cases} P = 42 \\ k = 1 \end{cases}$

$\begin{cases} \gamma = 5^1 \bmod 53 = \underline{5} \\ \delta = [(42 - 19 \cdot 5)1] \bmod 52 = \underline{51} \end{cases}$

(f) $VER_k(P, \gamma, \delta)$

è vera se $\beta^\gamma \gamma^\delta \equiv \alpha^P \pmod p$

$\beta^\gamma \gamma^\delta \bmod p = 35^5 5^{51} \bmod 53 = 35^5 (5^{10 \cdot 5} \cdot 5) \bmod 53 =$

$= 41 (4^{5 \cdot 5}) \bmod 53 = \underline{40}$

$\alpha^P \bmod p = 5^{42} \bmod 53 = (5^{10 \cdot 4} \cdot 5^2) \bmod 53 = 4^4 \cdot 5^2 \bmod 53 = \underline{40}$

(*essendo $5^{10} \bmod 53 = 4$)

$\Rightarrow$ OK!